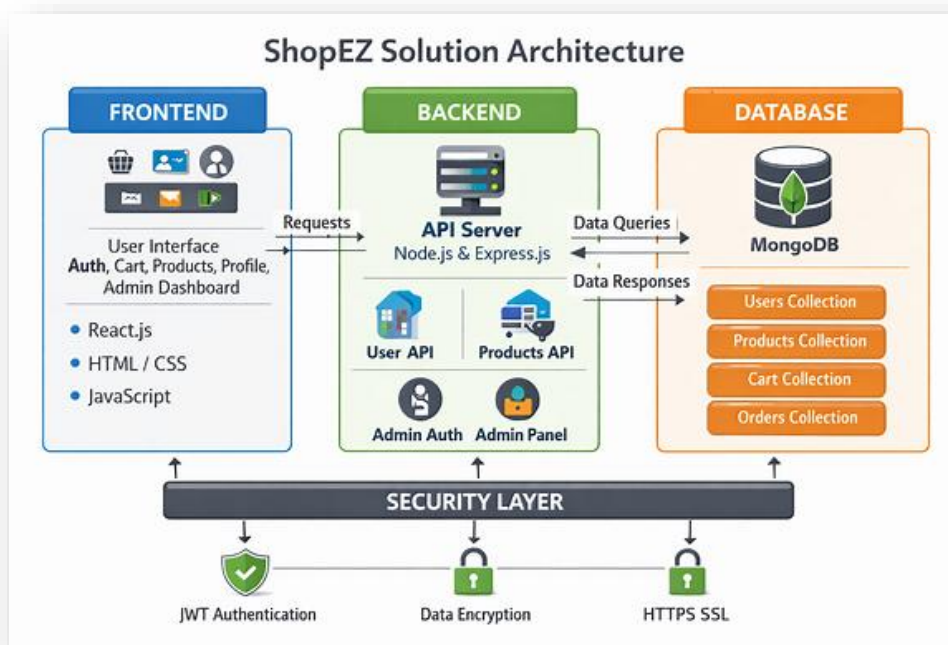


Project Design Phase Solution Architecture

Date	18 th June 2026
Student Name	Tirumala Srinidhi
Project Name	ShopEZ – E-Commerce Application
Maximum Marks	4 Marks

Solution Architecture:

The Shop-EZ Solution Architecture bridges the gap between business needs and technology implementation. It defines how various components — frontend, backend, database, and security — interact to deliver a seamless online shopping experience. The architecture ensures scalability, security, and efficiency for both users and sellers.



Goals

- Find the best tech solution for seamless online shopping.
- Define structure, behavior, and features for stakeholders.
- Provide clear specifications for development and delivery.

Architecture Layers

Layer	Components	Description
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Frontend	React.js, HTML, CSS, JavaScript	User interface with product catalog, cart, checkout, profile, admin dashboard.
Backend	Node.js, Express.js	API endpoints for users, products, cart, orders, and admin authentication.
Database	MongoDB	Collections for Users, Products, Cart, Orders, Admin.
Security	JWT, HTTPS, Encryption	Secure authentication, data protection, safe checkout.
Deployment	Cloud (Heroku/Netlify, MongoDB Atlas)	Scalable hosting, continuous integration, monitoring.

2. Objectives

- To design a robust and scalable e-commerce platform.
- To ensure secure transactions and data integrity.
- To provide real-time analytics for sellers and smooth navigation for buyers.
- To enable modular development, allowing future expansion and integration.

4. Data Flow

1. User Interaction:

- Users register/login via the frontend.
- Authentication handled through JWT tokens.

2. Product Browsing:

- Frontend requests product data from backend APIs.
- Backend fetches product details from MongoDB and returns JSON responses.

3. Cart & Checkout:

- Users add items to cart; cart data stored in MongoDB linked by user ID.
- Checkout triggers order creation and payment processing.

4. Order Confirmation:

- Backend sends confirmation to frontend.
- Users view order details in their profile section.

5. Admin Operations:

- Admin dashboard manages categories, banners, and product listings.
- Admin authentication ensures restricted access.

Development Phases

- **Phase 1:** Core frontend (catalog, cart, checkout) + backend APIs.
- **Phase 2:** User authentication, profile management, order history.
- **Phase 3:** Admin dashboard, analytics, seller tools.
- **Phase 4:** Scalability (cloud deployment, microservices).

Solution Requirements

- **Functional:** Secure checkout, product catalog, order tracking, admin tools.
- **Non-functional:** Scalability, performance, security, usability.
- **Business:** Commission-based revenue + premium seller subscriptions.

7. Key Benefits

- **User Experience:** Fast, intuitive, and secure shopping.
- **Seller Growth:** Analytics-driven insights for better decision-making.
- **Scalability:** Cloud-ready architecture supports future expansion.
- **Security:** End-to-end encryption and token-based authentication.